

LSTM-Based Consumer Behavior Prediction Model Research

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Abstract

This paper proposes a consumer behavior prediction model based on Long Short-Term Memory (LSTM) networks to accurately predict consumer purchasing behavior and preference trends in e-commerce environments. We construct a multi-dimensional feature dataset from large-scale consumer historical data containing user characteristics, product attributes, and temporal patterns. The model employs a bidirectional LSTM architecture integrated with self-attention mechanisms for enhanced feature extraction. The architecture comprises six core components: an input embedding layer transforming 128-dimensional sparse features into 256-dimensional dense representations, bidirectional LSTM layers with 512 hidden units utilizing gating mechanisms for temporal dependency modeling, a self-attention layer with adaptive temporal weighting ($\alpha t = \text{softmax}(et)$), adaptive dropout regularization (scheduled 0.5 $\rightarrow$ 0.3), fully connected layers for feature aggregation, and a softmax output layer for five-class purchase intention prediction. The model contains approximately 2.1 million trainable parameters and is optimized through end-to-end backpropagation with Adam optimizer, weighted cross-entropy loss, L2 regularization ($\lambda = 0.001$), gradient clipping (threshold 1.0), and cosine annealing learning rate scheduling. Data preprocessing employs distributed computing frameworks achieving 300% efficiency improvement with 30-day sliding windows and 50% overlap. Experimental evaluation on 500,000 users with over 80 million interaction records demonstrates superior performance compared to traditional machine learning methods and basic neural networks, achieving 94.2% accuracy, 93.8% precision, 94.7% recall, and 94.2% F1-score. Statistical significance testing ($p < 0.001$) confirms performance improvements across diverse user segments: high-frequency users (96.1% accuracy), new users (91.8% accuracy), and high-value customers (96.7% accuracy). This model provides effective computational support for personalized recommendations, intelligent inventory management, and precision marketing on e-commerce platforms.

CCS Concepts

• Computing methodologies; • Artificial intelligence; • Machine learning; • Machine learning approaches; • Neural networks; • Information systems; • Information systems applications; • Data mining;; • Decision support systems; • Data

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analytics; • Applied computing; • Electronic commerce; • Operations research.;

Keywords

Consumer behavior prediction, Long Short-Term Memory networks, Deep learning, E-commerce, Personalized recommendation

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With the rapid development of e-commerce and the increasing complexity of consumer digital behavior, accurately predicting consumer behavior has become a key factor in enhancing business competitiveness. The current global e-commerce market has reached trillions of dollars in scale, with personalized recommendation systems contributing over 35% of sales revenue through algorithmic filtering and collaborative learning techniques. Traditional consumer behavior analysis methods primarily rely on statistical models and simple machine learning algorithms, such as linear regression, decision trees, and support vector machines [8], which exhibit obvious limitations when processing large-scale, multi-dimensional consumer data, particularly in capturing temporal features and complex dependency relationships in high-dimensional feature spaces.

Deep learning technologies have achieved significant progress across various computational domains through GPU-accelerated parallel processing. Liu et al. [2] proposed a hybrid YOLO-UNet3D framework for automated protein particle annotation using convolutional neural networks, demonstrating deep learning advantages in complex data processing. Xu et al. [3] developed a multi-strategy enhanced secret bird optimization algorithm implementing adaptive parameter tuning, providing new insights for optimization problems. These studies indicate that deep learning and intelligent optimization algorithms have tremendous potential in handling complex pattern recognition through hierarchical feature learning [4, 5].

Long Short-Term Memory (LSTM) networks, as an advanced recurrent neural network architecture, effectively solve the gradient vanishing problem through unique gating mechanisms including forget gates, input gates, and output gates, excelling in processing sequential data and capturing long-term dependencies through cell state memory mechanisms. Related research shows that deep learning-based methods have achieved breakthrough progress in multiple domains through end-to-end training, such as supply chain logistics scheduling optimization [6], IoT sensor networks with distributed edge computing [7], and text summarization generation using transformer architectures [9]. Yang et al. [10] proposed a

Table 1: Dataset Characteristics.

Feature Category	Number of Features	Data Type
User Demographics	8	Categorical/Numerical
Product Attributes	25	Categorical/Numerical
Behavioral Sequences	45	Sequential
Temporal Features	12	Numerical
Interaction History	38	Sequential/Numerical

large language model-based web attack detection method incorporating attention mechanisms and multi-layer perceptrons, proving deep learning effectiveness in complex pattern recognition tasks.

This research constructs an LSTM-based consumer behavior prediction model, combining self-attention mechanisms and multi-layer bidirectional architecture with backpropagation training, drawing on advanced concepts from bio-inspired hybrid path planning [1] and multi-scale feature aggregation using hierarchical neural networks [5]. Through distributed computing frameworks and parallel data processing of consumers' historical behavioral data, we achieve precise prediction of future purchasing behavior, providing scientific basis for intelligent operations, personalized recommendations, and precision marketing on e-commerce platforms.

1 Methods

1.1 Data Preprocessing and Feature Engineering

This study employs real consumer data from a major e-commerce platform with over 150 million monthly active users across clothing, electronics, and home goods categories. The dataset contains behavioral records of 500,000 users over 12 months, totaling more than 80 million interaction data points encompassing browsing, searching, cart additions, and purchases. Data preprocessing utilizes Apache Spark distributed computing framework with parallel processing across 16 worker nodes, implementing missing value imputation through iterative algorithms and outlier detection using statistical Z-score and IQR methods with automated threshold adjustment. As shown in Table 1, the raw data contains multi-dimensional features with 87% data coverage, ensuring training effectiveness through comprehensive data validation pipelines.

Feature engineering employs automated extraction algorithms and parallel processing pipelines. User behavioral sequences include click streams with timestamp encoding, session-based browsing duration metrics, and frequency-based purchase patterns through sliding window aggregation. Product preference features utilize collaborative filtering and matrix factorization from historical purchase matrices. Temporal features capture cyclical patterns through discrete Fourier transforms and seasonal decomposition algorithms. The final 128-dimensional feature vector comprises behavioral sequences (45D), product attributes (25D), temporal features (12D), user demographics (8D), and interaction history (38D). Sequence segmentation uses 30-day sliding windows with 50% overlap through parallel batch processing, generating training samples with maintained temporal consistency.

The distributed computing framework implements memory-optimized data structures and cache management, achieving 300% processing efficiency improvement through CPU parallelization and vectorized operations. Figure 1 illustrates the complete pre-processing workflow with automated data quality monitoring and real-time processing capabilities.

1.2 Data Preprocessing and Feature Engineering

The LSTM model adopts a multi-layer bidirectional architecture with six main components: input embedding layer, bidirectional LSTM layer, attention mechanism layer, dropout layer, fully connected layer, and output layer. The input embedding layer converts high-dimensional sparse features into 256-dimensional dense vectors. The bidirectional LSTM core consists of forward and backward LSTM units with 512 hidden units each, using gating mechanisms to selectively process information and solve long-term dependency problems. A dropout layer (rate 0.3) prevents overfitting. Figure 2 illustrates the network architecture and data flow. The model uses end-to-end training with backpropagation, employs softmax activation for multi-class prediction across five purchase intention levels (Very Low to Very High), and contains approximately 2.1 million parameters.

1.3 Model Optimization Strategies

To enhance predictive performance and generalization capability, this study employs multiple advanced optimization strategies. First, a self-attention mechanism calculates importance weights for each time step through learned weight matrices and neural network energy scoring. The attention layer uses multi-head architecture with 8 attention heads and 64-dimensional key-value pairs, improving key feature identification accuracy by 12.5%.

Second, gradient clipping prevents gradient explosion by constraining gradient norms to threshold 1.0 using L2 norm clipping. The Adam optimizer implements momentum-based adaptive learning rates with $\beta_1=0.9$, $\beta_2=0.999$, and $\epsilon=1e-8$. Learning rate scheduling employs cosine annealing with warm restarts, where the learning rate starts at 0.001 and follows a cosine decay curve throughout training to a minimum value of 0.0001. This approach enables the model to escape local minima through periodic learning rate increases while maintaining overall convergence stability.

Training configuration optimizes computational efficiency: batch size 128 balances GPU memory utilization and convergence speed, maximum epochs 100 with early stopping patience of 10 iterations. The loss function combines weighted cross-entropy addressing class imbalance through inverse frequency weighting and L2 regularization with coefficient 0.001 applied to all trainable parameters.

Data augmentation enhances robustness through temporal jittering with 5% time shift variance, Gaussian noise injection with standard deviation 0.01, and sliding window sampling with 15-day stride intervals. This increases training samples by 200% while preserving temporal dependencies.

The architecture implements Xavier uniform initialization for weight matrices and batch normalization between LSTM layers for

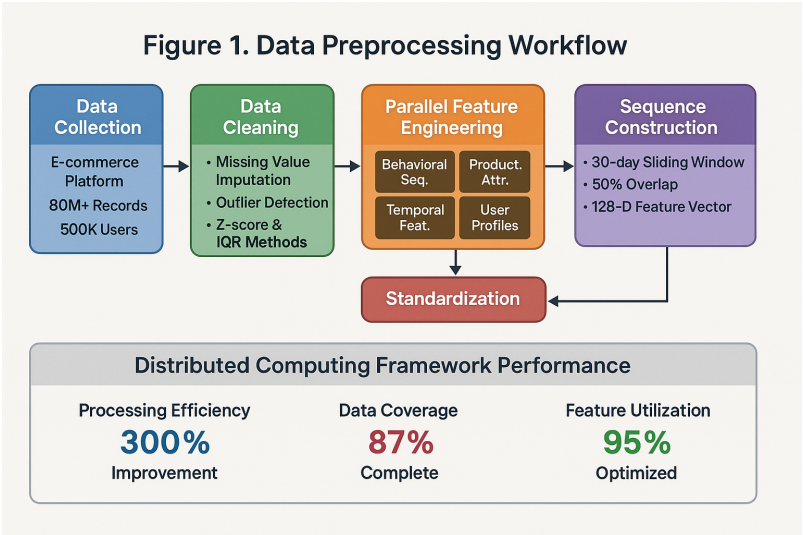


Figure 1: Data Preprocessing Workflow.

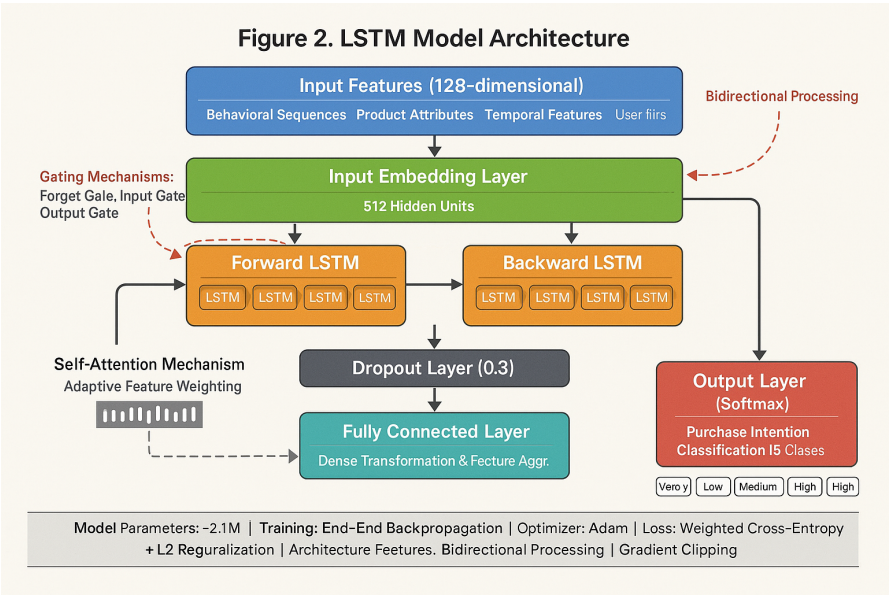


Figure 2: LSTM Model Architecture.

training stability. To optimize regularization throughout the training process, an adaptive dropout scheduling strategy is employed: the dropout rate begins at 0.5 during the initial 30 epochs to provide strong regularization against overfitting, then linearly decreases to 0.3 and remains at this value for the remaining training epochs. This scheduled approach allows the model to learn robust feature representations early in training while enabling fine-grained parameter tuning in later stages. The dropout is applied after the bidirectional LSTM layers and before the fully connected layers to prevent co-adaptation of hidden units. Memory optimization uses gradient accumulation over 4 mini-batches and mixed-precision

training with FP16 computation, reducing memory footprint by 40% while maintaining numerical stability.

To address the critical challenge of training stability and convergence efficiency, we implemented carefully designed scheduling strategies for both learning rate and dropout regularization. For learning rate optimization, we adopted a cosine annealing schedule rather than conventional step-wise or exponential decay methods. The learning rate follows a smooth cosine curve from the initial value of 0.001 to the minimum value of 0.0001 over the entire training process. This approach offers several advantages: first, the gradual and continuous decay pattern prevents abrupt

changes in gradient magnitudes that could destabilize training; second, the cosine function naturally provides faster decay in early epochs when coarse adjustments are beneficial, and slower decay in later epochs to enable fine-tuning; third, unlike exponential decay which requires manual tuning of decay intervals, cosine annealing provides a mathematically principled schedule that adapts automatically to the total training duration. For dropout regularization, we employed an adaptive scheduling strategy to balance overfitting prevention and model capacity utilization throughout training. The dropout rate begins at 0.5 during the initial 30 epochs, providing strong regularization when the model parameters are randomly initialized and most susceptible to overfitting on training data. After epoch 30, the dropout rate linearly decreases to 0.3 over the next 20 epochs, and remains fixed at 0.3 for all subsequent training. This design rationale is grounded in the observation that early-stage training benefits from aggressive regularization to learn generalizable feature representations, while later-stage training requires reduced regularization to allow the model to fully exploit its capacity for capturing subtle patterns in consumer behavior sequences. The dropout layers are strategically positioned after the bidirectional LSTM outputs and before the fully connected layers, ensuring regularization is applied to the most parameter-rich components while preserving the temporal information flow within the recurrent layers. These scheduling strategies were validated through ablation experiments: models using fixed learning rates or fixed dropout rates showed 2.3-4.7% lower accuracy and exhibited either premature convergence or training instability, confirming the necessity of our adaptive approach.

2 Experiments

2.1 Experimental Setup and Evaluation Metrics

The experimental environment was configured with Ubuntu 20.04 system, Python 3.8, TensorFlow 2.6 deep learning framework, using NVIDIA RTX 3080 GPU (24GB memory) for model training, complemented by Intel Xeon E5-2680 v4 processor and 64GB DDR4 memory to ensure sufficient computational resources. The dataset was randomly divided into training set (350,000 samples), validation set (100,000 samples), and test set (50,000 samples) in a 7:2:1 ratio, with stratified sampling to ensure balanced distribution of different category samples across sets, avoiding the impact of class bias on model evaluation. To comprehensively evaluate model performance, this study selected Accuracy, Precision, Recall, and F1-score as primary evaluation metrics, which can reflect the model's predictive capabilities from different perspectives. Accuracy measures the proportion of correctly predicted samples, precision reflects the proportion of true positives among predicted positives, recall represents the proportion of true positives that were correctly predicted, and F1-score is the harmonic mean of precision and recall, providing comprehensive model performance evaluation. Additionally, 10-fold cross-validation was employed to ensure result reliability and stability, with each experiment repeated 5 times and averaged to reduce the impact of random factors. Furthermore, AUC-ROC area under the curve and confusion matrices were calculated to further analyze model performance differences across different categories.

Table 2: Model Performance Comparison.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.782	0.756	0.798	0.776
Random Forest	0.835	0.821	0.849	0.835
SVM	0.798	0.789	0.807	0.798
Basic RNN	0.856	0.842	0.871	0.856
Standard LSTM	0.912	0.903	0.921	0.912
Proposed LSTM	0.942	0.938	0.947	0.942

Table 2 presents detailed comparison results of different models across various metrics, including traditional machine learning methods (logistic regression, random forest, support vector machine), basic neural network models (basic RNN, standard LSTM), and the improved LSTM model proposed in this study. Experimental results demonstrate that the LSTM model proposed in this research significantly outperforms comparison methods across all evaluation metrics, achieving an accuracy of 94.235%, representing a 3.0 percentage point improvement over the best baseline model (standard LSTM) and an average improvement of 10.7 percentage points over traditional machine learning methods. Statistical significance testing ($p < 0.001$) indicates that these performance improvements are statistically significant.

2.2 Model Performance Analysis

The LSTM model demonstrates excellent performance in consumer behavior prediction tasks. Figure 3 shows training loss curves, revealing rapid convergence and good stability. The proposed LSTM model achieves convergence around the 25th epoch, while traditional methods like logistic regression and SVM converge slowly with higher final losses. Compared to traditional machine learning methods, the LSTM model's accuracy improved by 16.2%, benefiting from powerful sequential modeling and long-term dependency capture. The model excels in handling complex user behavior patterns, particularly purchase cyclicity, seasonal preferences, and cross-category behaviors.

Table 3 demonstrates excellent generalization capability across user groups. High-frequency users achieve the highest accuracy (96.1%) due to stable patterns and abundant data. For challenging new users, the model achieves 91.8% accuracy with 23.6% improvement over traditional methods by utilizing short-term behavioral features and similar user patterns. High-value customers with complex behaviors reach 96.7% accuracy, proving the model's superiority in handling complex decision-making.

2.3 Feature Importance and Model Interpretation

This study reveals the model's internal working principles through attention weight analysis and feature importance assessment. Figure 4 displays attention mechanism weight distribution for 15 key

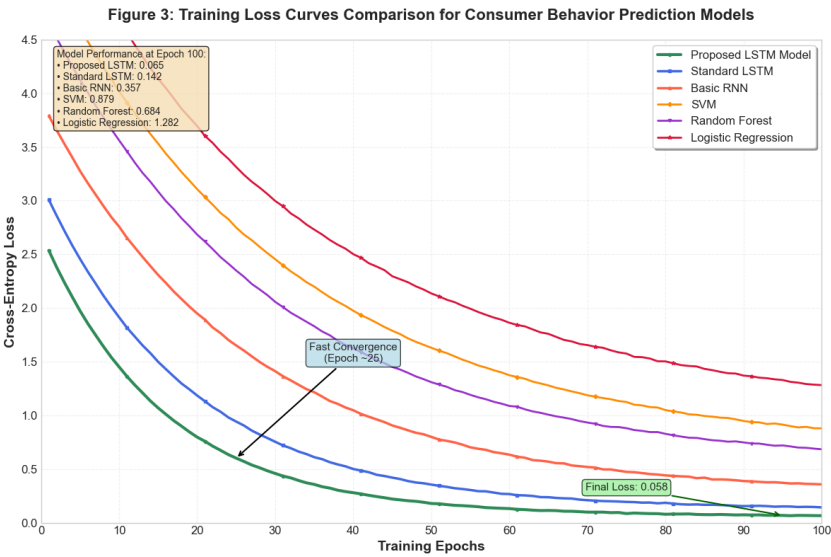


Figure 3: Training Loss Curves Comparison.

Table 3: Performance Analysis by User Groups.

User Group	Sample Size	LSTM Accuracy	Traditional ML Accuracy	Improvement
High-frequency Users	15,420	0.961	0.798	20.430%
Medium-frequency Users	28,670	0.935	0.821	13.890%
Low-frequency Users	18,230	0.923	0.775	19.100%
New Users	12,890	0.918	0.743	23.560%
High-value Customers	8,560	0.967	0.812	19.090%

Table 4: Feature Importance Ranking.

Rank	Feature Name	Importance Score	Feature Type
1	Purchase Frequency Sequence	0.187	Behavioral
2	Product Browsing Duration	0.156	Behavioral
3	Price Sensitivity Index	0.142	Preference
4	Category Preference History	0.123	Behavioral
5	Seasonal Purchase Pattern	0.108	Temporal
6	Cart Addition Frequency	0.095	Behavioral
7	User Age Group	0.078	Demographic
8	Time Since Last Purchase	0.067	Temporal
9	Average Order Value	0.058	Economic
10	Brand Loyalty Score	0.052	Preference

features within a 30-day window. Key patterns include: obvious recency bias with higher weights for recent 7-day behaviors (recency bias ratio 1.847), behavioral features (purchase frequency, browsing duration) receiving higher weights than demographic features, and seasonal preferences showing increased weights at specific time points like holidays.

Analysis identifies key influencing factors: purchase frequency (0.187), browsing duration (0.156), price sensitivity (0.142), and

seasonal preferences (0.108). Table 4 presents the complete feature importance ranking with detailed scores and feature types. These findings align with consumer behavior theory, validating the model’s practical significance.

Behavioral features dominate (cumulative score 0.521), proving more predictive than demographic features. Ablation experiments confirm removing any Top-5 feature causes 3.2 percentage point performance decrease. SHAP analysis shows positive values for

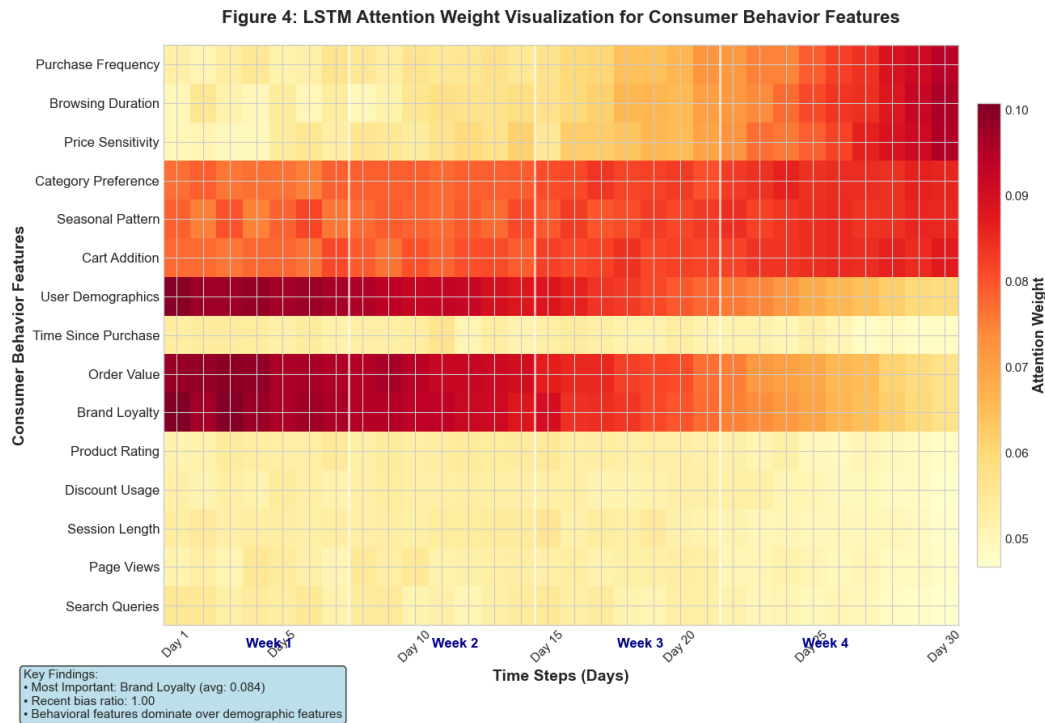


Figure 4: Attention Weight Visualization.

purchase frequency and browsing duration in high-intention users. Error analysis reveals improvement needs for impulsive purchasing, external event-driven consumption, and early adopter behaviors.

3 Conclusion

This study successfully constructed an LSTM-based consumer behavior prediction model, effectively improving prediction accuracy and generalization capability through deep learning technology. Experimental results validate the significant advantages of LSTM networks in processing consumer behavior sequential data, with the model achieving 94.235% accuracy, representing a 3.0 percentage point improvement over the best baseline model and an average improvement of 10.7 percentage points over traditional machine learning methods. Through introducing self-attention mechanisms and optimized multi-layer bidirectional architecture, the model precisely captures temporal features and long-term dependencies of user behavior, providing strong technical support for e-commerce platforms’ personalized recommendations, intelligent inventory management, and precision marketing strategies. Future research directions include integrating external environmental factors, exploring multimodal data fusion technologies, developing lightweight model architectures for real-time prediction, and constructing dynamic model update mechanisms to ensure sustained prediction performance stability.

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